HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F70LB

Life Insurance Mathematics B

Semester 2 2021/22

Duration: Two hours

You should answer ALL 8 questions.

Total 60 marks.

This is an open book exam but all answers must be in your own words to demonstrate your understanding. You must also show your working.

Answers that do not demonstrate understanding or show working may not receive marks.

Formulae and Tables for Examinations 2002 may be used. You may use a permitted calculator.

Upload time is for the purpose of scanning and uploading your answers. You must not use that time to continue working on your answers.

Question 1

Calculate the following:

(a) $\mu_{55:\,50}$

(1 mark)

(b) The probability that exactly one, but not both, of a life age 50 and a life age 60 die in the next 10 years.

(2 marks)

(c) $a^{(4)}_{55|50}$

(2 marks)

Basis: *PFA92C20 (female) and PMA92C20 (male) with 4% interest*
Assume the first life is female and second life is male.
You may assume that mortality of the lives are independent of each other.

[Total 5 marks]

Question 2

We consider two lives (x) and (y). Throughout this question, x refers to the first life and y to the second life. Describe in words what each of the following integrals represents and write down the corresponding international actuarial notation.

(a) $\int_0^{10} e^{-\delta t}\ {}_tp_{xy}\mu_{x+t}dt$

(2 marks)

(b) $\int_0^{20}(1 - {}_tp_y)\ {}_tp_x\ \mu_{x+t}dt$

(2 marks)

[Total 4 marks]

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Question 3

A life office sells with-profit whole-life insurance policies to lives aged 50. A sum assured of *£1000* is paid immediately on death. Level premiums are paid continuously throughout the term of the policy. The force of interest is δ and the force of mortality is μ_{x+t}.

(a) State Thiele's differential equation for this policy assuming no bonuses are awarded and no expenses.

(1 mark)

(b) Sketch a graph that shows both the net premium policy values and the gross premium policy values.

Basis
No expenses.
One bonus of £1000 is awarded at time 10, ie at age 60. This bonus increases the sum assured to £2000 from time 10 onwards.
The gross premium has been calculated using the principle of equivalence.
The same assumptions for mortality and interest are used for all premium and policy value calculations.

(3 marks)

(c) Suppose that premiums are paid at the start of each year, instead of continuously, and death benefits are paid at the end of the year, not immediately. Explain why you would use a recursive relationship between policy values, not linear interpolation, to calculate policy values at non-integer times.

(1 mark)

(d) Explain briefly how with-profit policies can be used by insurers to manage the risk of future changes in mortality.

(1 mark)

[Total: 6 marks]

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Question 4

A life insurer allows race-horse owners to buy 5-year term assurance policies in respect of their horses. It sells two types of policy, Policy 1 and Policy 2, each with a different policy design. Both policies were issued at the same time in respect of a population of 800 5-year-old horses each.

- Policy 1 has a single premium paid at the outset and more detailed underwriting than Policy 2.
- Policy 2 has level monthly premiums and a lower sum assured.

The following table shows the mortality of a standard population of race horses and the population and deaths for each of the two types of policy. The population data in columns (A) show the remaining number of policies in force at the start of each year.

	Standard population	Policy 1			Policy 2		
Age	Death rate	(A) Population	(B) Deaths	(C) = (B)/(A)	(A) Population	(B) Deaths	(C) = (B)/(A)
5	1.0%	800	4	0.5%	800	8	1.0%
6	2.0%	796	15	1.9%	600	14	2.3%
7	3.0%	781	24	3.1%	520	18	3.5%
8	4.0%	757	28	3.7%	360	16	4.4%
9	5.0%	729	35	4.8%	200	12	6.0%
Total		**3,863**	**106**	**2.7%**	**2,480**	**68**	**2.7%**

(a) Calculate the standardised mortality ratio (SMR) for Policy 1 and Policy 2 compared to the standard population. (2 marks)

(b) Explain why the crude death rate (CDR) is the same for both policies even though death rates have been higher every year under Policy 2. (2 marks)

(c) List, with brief reasons, 3 types of selection which appear to be present in the table above and explain what might have caused each of them. (6 marks)

[Total 10 marks]

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Question 5

An insurer issues 3-year unit-linked policies with the following features.

Policyholder age at start of policy: 47 exact
Sum assured on death: £1000 plus the bid value of the unit fund, paid at the end of year of death
Premiums: £1000 at the start of each year

Allocation factor: 0.95 in year 1, 1.00 in year 2, 1.01 in year 3
Bid\offer spread: 2%
Fund management charge: 20% of the return on the fund that year plus 2% of the total year-end fund, paid at the end of each year

Basis
Mortality: ELT15 (Females)
Unit fund rate of return: 3% in the first year, 5% in the second year, 10% in the third year
Sterling fund rate of return: 2% a year
Initial expenses: £100 at the start of year 1
Renewal expenses: £10 at the start of year 2 and year 3
Fund management expenses: 1.5% of the fund at year end before fund management charge

(a) Calculate the profit vector and profit signature for this policy.

(8 marks)

(b) Suggest, with reasons, which feature of this policy is most likely to result in profit being higher or lower than expected.

(2 marks)

The insurer decides to allow surrenders in this policy:

- 20% of policyholders surrender their policies at the end of the first year (after all charges) with a penalty of 10% of the bid value of the units
- 50% of policyholders whose policies are in force at the start of the second year surrender their policies at the end of the second year (after all charges) with a penalty of 5% of the bid value of the units

(c) Calculate the profit signature allowing for surrenders.

(4 marks)

[Total 14 marks]

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Question 6

(a) Explain how the risks to the member are different in a defined-benefit (DB) pension scheme compared to a defined-contribution (DC) pension scheme.

(2 marks)

A pension scheme member is age 55 exact. Her salary is currently £60,000. She has 10 years of service in her pension scheme.

(b) Calculate the past service liability.

Scheme benefits
Normal retirement age is 60.
Accrual rate is one-sixtieth.
Final Pensionable Salary (FPS) is the average of earnings in the last 1 year before retirement or earlier exit.
Members who withdraw before age 60 receive a deferred pension which comes into payment from age 60. Deferred pensions increase by 2% for each complete year of deferment.
There are no dependants' benefits and no ill-health pensions.
Pensions are paid monthly in advance.
Pensions do not increase in payment.

Basis
Salaries increase by 3% at the end of each year
Withdrawals occur in the middle of the year
Retirements occurs at exact age 60
Gross interest rate 4%
Mortality in retirement PFA92C20
No mortality before age 60

Service table (where r(x) is exits through age retirement and w(x) are exits through withdrawal):

Age	l(x)	r(x)	w(x)
55	1000	0	0
56	1000	0	0
57	1000	0	0
58	1000	0	50
59	950	0	100
60	850	850	0

(9 marks)

[Total 11 marks]

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Question 7

An insurer provides the following information about one of its policies and the assumptions it uses for profit-testing.

Time (t)	$p_{x+(t-1)}$	Cashflows at time t before allowing for reserves	Profit signature with no allowance for reserves	Reserve at time t-1
1	0.99	-100	-100.00	0
2	0.99	50	49.50	0
3	0.95	100	98.01	0
4	0.95	500	465.55	500
5	0.95	-500	-442.27	500

Interest on reserves is 3% and the risk discount rate is 12%. The expected value of profit before allowing for reserves is £64.84.

(a) Calculate the profit signature and expected present value of profit with allowance for reserves.

(4 marks)

(b) Without doing any further calculations, say what the expected present value of profit would be if the interest on reserves is 12% instead of 3%. Give reasons for your answer.

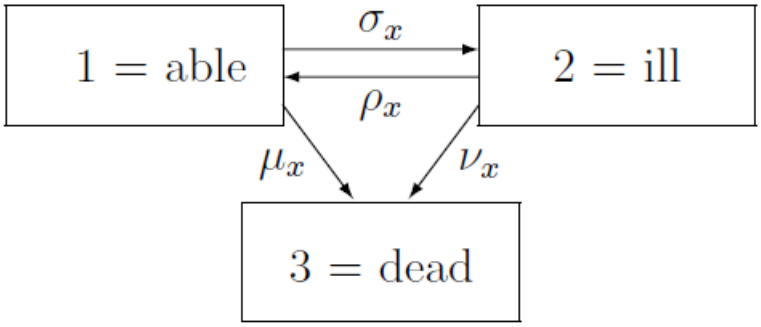
(2 marks)

[Total 6 marks]

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Question 8

An insurer offers Income Protection Insurance (IPI) policies. The insurer uses the following simple Markov model to price its policies.



Describe in words what each of the following integrals represents.

(a) $50{,}000 \int_1^{\infty} {}_tp_x^{\overline{11}} \mu_{x+t} e^{-\delta t} dt$

(2 marks)

(b) $10{,}000 \int_0^{65-x} {}_tp_x^{\overline{11}} \sigma_{x+t} e^{-\delta t} \int_0^{65-(x+t)} {}_sp_{x+t}^{\overline{22}} e^{-\delta s} ds\, dt$

(2 marks)

[Total 4 marks]

END OF PAPER